

# VARUN SINGHAL

San Diego, CA | (858) 444-0091 | varunraisinghal12@gmail.com | linkedin.com/in/varunraisinghal | github.com/varunraisinghal

## EDUCATION

**University of Southern California** Los Angeles, CA  
**Master of Science in Computer Science: Artificial Intelligence** January 2025 – December 2026  
**Concentration Focus:** Artificial Intelligence, Machine Learning, Quantum Computing + Cryptography

**University of California, San Diego** San Diego, CA  
**Bachelor of Science: Computer Science** October 2020 – June 2024  
**GPA:** 3.83/4.00, Provost Honors (2020 – 2024)

## PROFESSIONAL EXPERIENCE

**DES Labs** Los Angeles, CA  
**Co-Founder** 2025 – Present

- Built a proprietary **OCR and field-extraction** pipeline for purchase agreements, disclosures, and listing documents, auto-filling the fields real estate agents previously retyped from PDFs.
- Deployed with **2 real estate clients** on 100 documents per type, cutting agent paperwork **from hours to minutes**.

**GEICO** San Francisco, CA  
**Software Engineer** May 2024 – August 2025

- Optimized a production quote orchestration and estimation system, reducing **end-to-end latency from ~30s to <13s** and **compute costs by 50%+** while meeting a <15s SLA and preserving quote consistency.
- Re-architected the **critical path** by eliminating unnecessary service calls, **parallelizing independent requests**, reducing duplicate computation, and deferring non-critical processing that did not affect initial pricing decisions.
- Led a cross-team **ML scoring-model initiative** from feature design through production integration to improve premium-estimation accuracy across partner quoting workflows.
- Productionized ML and backend services using **AWS SageMaker, Lambda, DynamoDB, REST APIs, and CI/CD**, improving reliability and supporting high-volume production inference.

**Amazon Web Services (AWS)** Seattle, WA  
**Software Engineering Intern** May 2023 – August 2023

- Worked directly with **Fortune 500** customers to design and provision production database and storage pipelines, integrating new workloads with existing enterprise environments and **onboarding 5 customer workloads**.
- Built automated **CloudWatch and GuardDuty** health checks that reduced failure-detection time by **~40%**.

## RESEARCH & TECHNICAL PROJECTS

**VL4M4Rec: Multimodal Semantic Representation for Recommendation with Large Vision-Language Models** arXiv, March 2026  
*Python, PyTorch, LLaVA-NeXT 7B, Sentence-BERT, CLIP, Hugging Face, NVIDIA GPU*

- Built an end-to-end **multimodal recommender system** using LLaVA-NeXT 7B semantic descriptions, Sentence-BERT item embeddings, and cosine-similarity retrieval over user histories.
- Designed and optimized a GPU inference pipeline using **4-bit quantized LLaVA-NeXT 7B** on an **NVIDIA RTX 5080**, batching 4,708 catalog items and precomputing/caching semantic embeddings to shift expensive inference offline.
- Evaluated **12 text, vision, and multimodal fusion approaches** across 2,914 users; achieved **0.3539 Recall@10** and **0.2745 NDCG@10**, improving Recall@10 by **54.9%** over the BERT baseline.

**Low Latency C++ Arbitrage Engine | C++**

- Implemented a graph-based **C++** arbitrage engine and real-time market-data pipeline across Binance, Coinbase, and Kraken with **<10ms detection** and **<300ms execution latency**.

**Prediction Market Arbitrage Scanner | Python, FastAPI**

- Built a **real-time** pipeline to ingest, normalize, semantically match, and validate Polymarket/Kalshi markets; event-level filtering reduced **LLM calls** by **~80%** while minimizing false-positive matches.

## RELEVANT SKILLS

- Languages:** C/C++, Python, SQL
- GPU & AI:** CUDA C/C++, CUDA Kernels, Parallel Programming, GPU Programming, NVIDIA GPUs, NVIDIA Nsight, Machine Learning, Deep Learning, AI Algorithms, PyTorch, TensorFlow, Hugging Face Transformers, LLaVA-NeXT, Sentence-BERT, CLIP, Vision-Language Models, Recommender Systems, Computer Vision, NLP
- Systems & Performance:** Linux, Computer Architecture, Software Design, Algorithms, Low-Level Performance Optimization, Model Inference, Quantization, Information Retrieval, Latency & Throughput Optimization, Performance Profiling, AWS SageMaker, CI/CD